



Kind Attn. – Mr. Murli Tiwari
M/S. Aditya Birla

Project location-Renukoot, India
Date- 29/07/2022
Ref no- DA/EC/AB/AT/974

Subject - Proposal for Audiometric Booth

Dear Sir,

ABOUT US

We at DECIBELS ACOUSTICS offers Tailor-made solutions for all acoustic and thermal insulation needs, our vast experience makes us most preferred choice for our customers in this industry, with more than 25 plus years of combined experience we have been at forefront of acoustic design and analysis and executed complex projects at both small and large scales. Our passionate team works on minute detailing and offer best fit solutions both considering budget and technical.

What we offer

-  Industrial acoustic solutions
-  Plant noise and thermal acoustic solutions
-  Traffic noise control
-  Architectural Acoustic solutions
-  Audiology testing solutions
-  Acoustic consultancy

Requirement-

Audiometric Booth for ENT testing of size **4 feet x 4 feet x 7 feet** (LxBxH) and **5 feet x 5 feet x 7 feet** (LxBxH) (outer dimensions) respectively, for around 40 dB(A) inside the chamber from an ambient of 60-65 dB(A) as per **OSHA norms**.



Solution -

Structure – (Chamber structure details)

Entire outer structure/panels shall be modular without base frame made from CRCA sheet of 1.6mm of TATA/JINDAL. Panels thickness for side walls shall be 100mm and roof panels shall be of 100mm thickness. Cross members at 600/600 C/C shall be providing within the panels for structural stability and strength. Inner side of the panels shall consist of with GI perforated sheet of 0.4mm thickness. Entire enclosure design shall be adopted for maximum rigidity and environment performances, all panel to panel and base to panel joints shall be optimized with thin rubber acoustic seals. Structure shall be mounted with L shape steel plates with anchor fasteners. Corner joints shall be covered with MS coated L shape profiles with the help of self-tapping GI screws.

Acoustic Insulation

Panels shall be insulated with combination of acoustic dampeners and acoustic absorbers to achieve desired level of acoustic performance through the frequency range.

Side Panels- (4 inches thick)

A layer of 3mm thick acoustic damper with very high density of 2000kg/m³ shall be adhered on the inner surface of the panel, 100mm thick resin bond mineral wool slab (PROROX SL960) confirming IS-8183 with density 100kg/m³ should be covered with 50 GSM glass fibre tissue paper over that 0.4mm thick perforated G.I sheet shall be self-screwed which will provide the perforation of 40-42% for heavy insulation at last acoustic fabric of customer choice colour shall be pasted for aesthetic look. Total thickness of the side panels shall be of 100 mm.

Roof Panels- (4 inches thick)

A layer of 3mm thick acoustic damper with very high density of 2000kg/m³ shall be adhered on the inner surface of the panel, 100mm thick resin bond mineral wool slab (PROROX SL960) confirming IS-8183 with density 100kg/m³ should be covered with 50 GSM glass fibre tissue paper over that 0.4mm thick perforated G.I sheet shall be self-screwed which will provide the perforation of 40-42% for heavy insulation at last acoustic fabric of customer choice colour shall be pasted for aesthetic look. Total thickness of the side panels shall be of 100 mm and one nos of ceiling light shall be provided.

Acoustic doors

Acoustic doors with similar insulation shall be provided as per final approved drawing from the client. Doors shall be fixed with heavy duty hinges and locks and periphery of the door shall be fixed with EPDM gaskets to avoid sound leakage.



Acoustic Dampener

High density polymer-based, asphalt-free, synthetic soundproofing membrane that offers good acoustic insulation in different building elements. It is equipped with a tissue non-tissue polypropylene fleece on its upper side to improve the tear strength of the membrane and at the same time offer a protection or a finishing.

Acoustic Absorber

A nature-based insulation made of volcanic stone, a safe and sustainable material that continues to perform and lasts throughout the lifetime of the building. It is designed and manufactured to achieve excellent acoustic insulation performance in all types of drywall partition configurations.

Vision Panel

A single unit of sound proof glass (DGU) shall be provided which shall be made of 6mm clear toughen on both sides and 12mm aluminum bidding with air gap shall be provided total thickness will be 24mm.

Hardware

Zinc passivity hardware shall be used for fixing the chamber, and hinges shall be fixed as per the specification provided by the client.

Jack Panel

8-10 nos of patch chords having dia 6.5mm shall be provided for the connectivity of headphones and audiometer for testing at different frequency.

Ventilation system

Forced air ventilation system shall be provided as per the HVAC calculation with the help of duct and silent fan.

Finish

Entire outer surface shall be chemically cleaned before red-oxide, Enamel paint of desired RAL colour shall be done on the entire outer surface of the booth for better finish and on inner side of booth the fabric finish shall be provided for better aesthetic.



Commercials

Sl. No.	Description	QTY	Rate	Amount
1.	Fabrication and installation of Audiometric booth as per above mentioned specification(4x4x7) feet.	1 nos	1,20,000/-	1,20,000 INR
2.	Fabrication and Installation of Audiometric Booth as per above mentioned specification(5x5x7) feet.	1 nos	1,30,000/-	1,30,000 INR

Result

Audiometric booth is designed to provide you less than 40dB (A) noise level inside the chamber at an ambient of 60-65 dB(A) outside the chamber as per OSHA norms, all the materials used are fire resistant and water repellent.

Terms and Conditions

1. Price Ex-Works, Noida transport cost will be extra in client scope
2. Payment Term- 50%, advance along with P.O, 40% on dispatch and 10% after installation/testing within seven working days.
3. GST @ 18% shall be charged extra.
4. Installation cost is included.
5. Unloading at site, Scaffolding, ladder, any other support for lifting man and material during installation shall be in client's Scope.
6. Electricity at site shall be provided by the client.
7. Unloading at site shall be in client scope.
8. Delivery time 2-3 weeks after approvals & advance payments

Hope this proposal will suit your requirement.

For Decibels Acoustics

Anil Thakur

+91- 9643427078, 9818093940